$$\frac{d\psi_0}{dx} = -\frac{\omega x}{\hbar}\psi_0(x).$$

This is a simple differential equation that you can easily solve. Or, you can just check that the ground-state wave function

$$e^{-\frac{\omega}{2\hbar}x^2}$$

in Eq. 10.15 solves it. Calculating the wave functions for the excited (nonground) states is even easier—we don't even have to solve any equations. Let's go up the ladder to $n = +1$. We can do that by applying $\mathbf{a}^+$ to the ground state. Let's call the wave function of this new state $\psi_1(x)$.

To avoid dragging the constant $-i/\sqrt{2\omega\hbar}$ around in our calculations, we'll just drop it in our definition of $\mathbf{a}^+$. This only affects the numerical coefficient. The resulting equation is

$$\psi_1(x) = (\mathbf{P} + i\omega\mathbf{X})\psi_0(x)$$

or

$$\psi_1(x) = \Big(-i\hbar\frac{\partial}{\partial x} + i\omega x\Big)e^{-\frac{\omega}{2\hbar}x^2}.$$

Factoring out the i, we get

$$\psi_1(x) = i\Big(-\hbar\frac{\partial}{\partial x} + \omega x\Big)e^{-\frac{\omega}{2\hbar}x^2}.$$

The "hardest" part of working this out is performing an easy derivative of $e^{-\frac{\omega}{2\hbar}x^2}$. Here is the result: